Why Warm Weather May Have Saved the Crocodiles

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Sixty-six million years ago, a huge asteroid hit Earth and wiped out the dinosaurs. But crocodiles survived. Why? New research says their love of warm weather may have saved them. Scientists led by University College London, or UCL, studied more than 7,000 **fossils** and records from over 400 living and **extinct species**. They used computer models to rebuild Earth's climate and geography over 245 million years. Their study was published in Science Advances. For much of their 250-million-year history, crocodile relatives followed warm climates. When Earth got hotter, they spread to new places, sometimes as far as Antarctica. When it got colder, they moved back toward the **tropics**. After the asteroid hit, computer models suggested that tropical places near the **equator** stayed warm enough to shelter crocodile relatives, even while life got harder for many dinosaurs. Only the most extreme models took away these safe places. Lead author Dr. Alfio Alessandro Chiarenza, of UCL Earth Sciences, said the asteroid did not make the whole planet equally hard to live in for every group.

Word Treasure

fossils -- Fossils are the remains or traces of plants and animals that lived long ago.
extinct -- Extinct means a kind of plant or animal no longer lives anywhere on Earth.
species -- A species is a group of living things that are alike.
tropics -- The tropics are the warm parts of Earth near the middle of the planet.
equator -- The equator is an imaginary line around the middle of Earth.

Background you need

Crocodiles are large reptiles that live in warm, wet places like rivers and swamps, where they can stay warm and find food.

When the climate changes, animals that need warmth often move to places that stay warm enough for them to live.

After a huge asteroid hit Earth, many dinosaurs died, but crocodile relatives in warm places near the equator could survive because those places stayed warm enough for them.

Break it down

WHO: Crocodile relatives and scientists led by University College London (UCL), including lead author Dr. Alfio Alessandro Chiarenza of UCL Earth Sciences

WHAT: Crocodile relatives survived the asteroid that wiped out the dinosaurs, and new research links their survival to their love of warm weather

WHERE: Tropical places near the equator that stayed warm after the asteroid hit, and, when Earth was hotter, places as far as Antarctica

WHY: Tropical places near the equator stayed warm enough to shelter crocodile relatives after the asteroid hit, even while life got harder for many dinosaurs

Why it matters

It shows how finding a safe, warm shelter can help living things survive big changes in their surroundings.

Test what you remember

1. What happened sixty-six million years ago?

(A) Crocodile relatives first appeared on Earth.
(B) A huge asteroid hit Earth and wiped out the dinosaurs.
(C) Earth became cold everywhere at the same time.
(D) Scientists published a study in Science Advances.

2. How many fossils and records did the scientists study?

(A) Exactly 245
(B) Fewer than 400
(C) More than 7,000 fossils and records
(D) More than 250 million

3. Who led the scientists in this research?

(A) Crocodile relatives
(B) Scientists led by University College London, or UCL
(C) A team from Antarctica
(D) The dinosaurs

4. When Earth got hotter, how far did crocodile relatives sometimes spread?

(A) Only near the equator
(B) Back toward the tropics
(C) As far as Antarctica
(D) Only to the tropics

5. What did the computer models suggest about tropical places near the equator after the asteroid hit?

(A) They became too cold for every group.
(B) They disappeared completely.
(C) They moved to Antarctica.
(D) They stayed warm enough to shelter crocodile relatives.

6. What did crocodile relatives do when Earth got colder?

(A) They stayed where they were.
(B) They spread to Antarctica.
(C) They moved back toward the tropics.
(D) They followed the dinosaurs.

Answer key (try the quiz first!): 1: B 2: C 3: B 4: C 5: D 6: C

Questions to think about

1. Lead author Dr. Alfio Alessandro Chiarenza: The asteroid did not make the whole planet equally hard to live in for every group.

MY THOUGHTS (at least 20 words):